

Varicose Veins of the Lower Extremity: Doppler US Evaluation Protocols, Patterns, and Pitfalls

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Abbreviations: AAGSV = anterior accessory GSV, CFV = common FV, DVT = deep vein thrombosis, ELA = endovenous laser ablation, FV = femoral vein, GSV = great saphenous vein, PV = popliteal vein, SSV = small saphenous vein

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SA-CME LEARNING OBJECTIVES

After completing this journal-based SA-CME activity, participants will be able to:

- Identify the various deep and superficial veins of the lower extremity, as well as common drainage patterns.
- Recognize imaging findings used to accurately diagnose deep and superficial venous insufficiency and the potential imaging pitfalls.
- Discuss complications associated with and treatment options for venous insufficiency.

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Venous insufficiency is a cause of substantial morbidity and medical expenditures. Diagnostic US evaluation of venous insufficiency requires a thorough understanding of the venous anatomy, including the deep, superficial, and perforator veins. The highly variable venous anatomy requires that operators use sound judgment to expand on protocol images and thus avoid missing important sources of reflux. The US examination requires specific patient positioning and use of provocative maneuvers. A basic understanding of the pathophysiology of venous insufficiency and the various treatment methods helps to identify key observations so that ineffective treatment methods are not pursued. The examination reports should have greater detail than those for the more common lower extremity deep venous thrombosis evaluation, requiring numeric and narrative descriptions of deep and superficial venous patency, reflux, diameter, and pathways. Potential pitfalls include not recognizing or detecting deep venous reflux, misidentifying common femoral vein reflux as deep venous reflux when the reflux is isolated or related to saphenofemoral insufficiency, not recognizing anterior accessory great saphenous vein (AAGSV) involvement in saphenofemoral insufficiency, not recognizing or reporting great saphenous vein or AAGSV superficialization, not suspecting central venous obstruction, and not realizing when provocative maneuvers were ineffective. With knowledge of the lower extremity venous anatomy, venous insufficiency pathophysiology, basic treatment strategies, protocol best practices, patterns of observation, and diagnostic pitfalls, those who interpret venous insufficiency US studies can perform examinations and deliver reports that help patients receive appropriate treatment.

Online supplemental material is available for this article.

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Introduction

Varicose veins and venous insufficiency are common, with varying prevalence rates; in one study (1), varicosities were found in 40% of men and 16% of women. In addition to causing varicosities, chronic venous insufficiency can cause eczema, edema, hyperpigmentation, and scarring, with more severe cases resulting in venous stasis ulcers (Figs E1, E2). These ulcers can cause significant morbidity, with up to 20% of ulcers persisting at 2 years and nearly two-thirds of patients experiencing episodes of ulceration that last longer than 5 years. Estimates of the economic effect of venous stasis disease suggest costs of 2 million working days and \$3 billion per year in the United States alone, with up to 3% of the total health care budget in developed countries spent on treatment of varicose veins and associated complications (2).

Successful treatment of venous insufficiency depends on an adequate diagnostic evaluation and appropriate selection of treatment. It is also important to identify those patients who are unlikely

TEACHING POINTS

- The SSV cranial extension courses superiorly and drains into the upper PV, the GSV through an oblique communication (typically the posterior thigh circumflex vein), or the gluteal or pudendal veins. Cranial extension can occur with or without coexisting drainage into the PV (with or without a saphenopopliteal junction).
- Numerous small variable perforator veins normally function to contribute to the drainage of the superficial veins into the deep venous system.
- For insufficiency studies, the legs must be positioned below the level of the patient's head to maximize venous filling and optimize evaluation for reflux; this is optimally achieved by having the patient lie on a tilt table in the reverse Trendelenburg position at a 60° or greater incline.
- Detailed reporting of an examination for venous insufficiency ensures that a patient is well suited for treatment, influences insurance reimbursement, and guides appropriate treatment.
- Clues to possible central obstruction include loss of normal phasicity in the CFV and the detection of unusual collaterals such as dilated superficial epigastric veins, pudendal veins, or upper posterior thigh veins.

to respond well to certain types of treatment. In this article, we review the lower extremity venous anatomy, pathophysiology of venous insufficiency, and treatment options for varicose veins as background for more detailed consideration of the Doppler US protocols, patterns, and pitfalls that are important in the diagnostic evaluation of venous insufficiency.

Lower Extremity Venous Anatomy

The lower extremity veins can be grouped into three categories: deep veins, superficial veins, and perforator veins.

Deep Veins

The deep veins of the lower extremity follow the major arteries and are named according to similar conventions. The deep veins of the calf are usually paired and include the anterior tibial vein, posterior tibial vein, and peroneal vein (Fig 1). The peroneal vein drains the posteromedial lower leg into the tibioperoneal trunk. The posterior tibial vein drains the posterior compartment of the leg and plantar surface of the foot into the tibioperoneal trunk (3). Deep veins of the thigh include the common femoral vein (CFV), deep femoral vein (FV), FV, and popliteal vein (PV) (Fig 1). The PV travels posteriorly, medial to the artery below the knee, superficial to the artery at the knee, and lateral to the artery above the knee. The PV becomes the FV at the adductor hiatus. The FV travels along the anteromedial thigh, medial to the artery. The FV is sometimes referred to as the superficial FV because it travels with the superficial

femoral artery; however, current best practice is to avoid use of the term *superficial* in reports, as this may be confusing, such that some think that the vein is not a deep thigh vein. The deep FV arises from the deep thigh, with the lower portions difficult to visualize on US images, and combines with the FV to become the CFV. The CFV crosses the inguinal ligament and becomes the external iliac vein (3).

Superficial Veins

Innumerable tiny venules in the subcutaneous tissues join to form minor veins that combine with other minor veins to eventually drain into two main lower extremity superficial veins known as the great saphenous vein (GSV) and small saphenous vein (SSV) (Figs 2, 3). The GSV classically arises from the medial aspect of the dorsal foot, ascends anteriorly to the medial malleolus, passes posteromedially to the knee, and ascends medially in the thigh. Drainage of the GSV into the CFV is characteristically at the groin crease at or above the area where the deep FV joins the FV. A few variations in the GSV anatomy have been described, although many are better described as large tributaries or accessory veins. Full-course duplication of the GSV is rare, seen in only about 1% of the population (3,4).

While the location of the saphenofemoral junction is relatively constant in the groin, the exact location and pathway of veins that drain into the GSV are highly variable. The anterior accessory GSV (AAGSV) is a common vein coursing through the lateral and anterior thigh to connect to the GSV near the saphenofemoral junction, with the posterior accessory GSV coursing through the posterior thigh to connect to the GSV (Figs 2A, 3). Other named veins may branch from these accessory veins or drain directly to the GSV, but specific labeling of these vessels is not clinically important.

For reference, the largest vein that travels obliquely through the anterior thigh anterolateral to the AAGSV, draining either to the AAGSV or separately to the GSV, can be labeled the anterior thigh circumflex vein, while the largest vein that travels obliquely through the posterior thigh posteromedial to the posterior accessory GSV and drains to the posterior accessory GSV or separately to the GSV can be labeled the posterior thigh circumflex vein. In the calf, the largest posterior tributary draining into the GSV below the knee has been called the posterior arch vein or posterior calf circumflex vein; the largest anterior GSV branch below the knee has been called the anterior calf circumflex vein or anterior tributary vein (Figs 2A, 3) (3).

The SSV classically arises from the lateral margin of the dorsum of the foot, ascends posterior

Figure 1. Photograph shows the lower extremity deep veins overlaid on a model in the supine position for evaluation of the lower extremity deep veins. The red line separating the femoral vein (FV) from the popliteal vein (PV) represents the adductor hiatus. ATV = anterior tibial vein, CFV = common FV, DFV = deep FV, PTV = posterior tibial vein, PeV = peroneal vein.

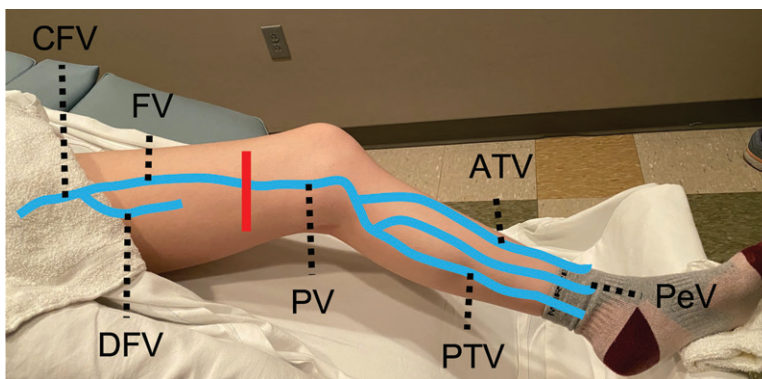
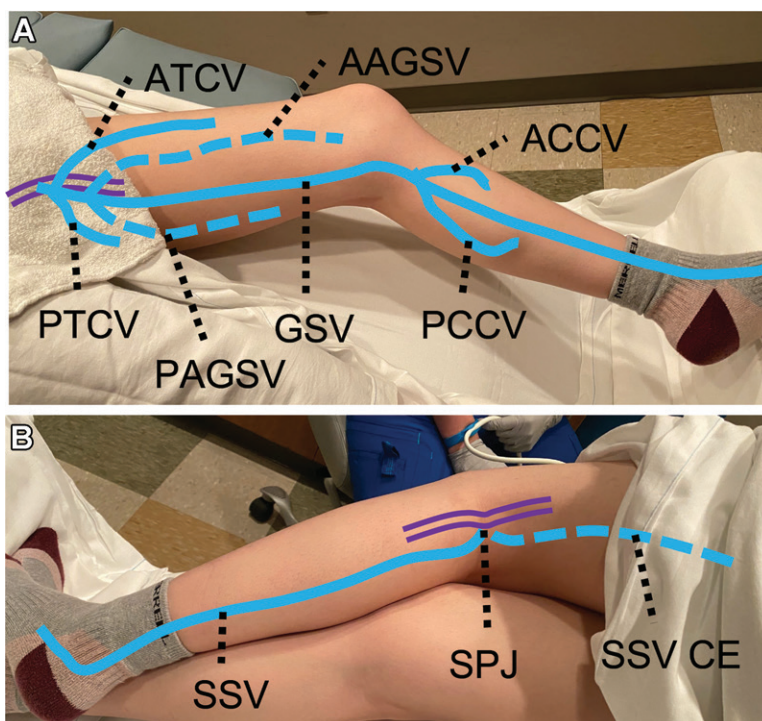


Figure 2. Photographs of lower extremity superficial veins overlaid on a model positioned for examination of the GSV (A) and SSV (B). (A) The model is in the reverse Trendelenburg position, with the hip externally rotated and the knee slightly flexed. The purple lines indicate the saphenofemoral junction and the CFV-FV. AAGSV = anterior accessory GSV, ACCV = anterior calf circumflex vein, ATCV = anterior thigh circumflex vein, PAGSV = posterior accessory GSV, PCCV = posterior calf circumflex vein, PTCV = posterior thigh circumflex vein. (B) The model is positioned in the right lateral decubitus position for examination of the left leg. The purple lines indicate the saphenopopliteal junction (SPJ) and PV. Cranial extension (CE) of the SSV also is seen. Note that the model is not positioned at the optimal angle of greater than 60° to optimize the image for labeling purposes.



to the lateral malleolus and posterolaterally in the calf, and is highly variable beyond this course (Figs 2B, 3). Classically, the SSV drains into the PV in the popliteal fossa. However, a substantial number of patients have variable anatomy, with the SSV sometimes draining into deep veins above the level of the PV, such as the FV, or draining to the GSV by way of an intersaphenous vein. A common and important variant is the cranial extension or thigh extension, also known as the vein of Giacomini (5), which was originally reported by Giacomini as being present in more than 70% of patients. The SSV cranial extension courses superiorly and drains into the upper PV, the GSV through an oblique communication (typically the posterior thigh circumflex vein), or the gluteal or pudendal veins. Cranial extension can occur with or without coexisting drainage into the PV (with or without a saphenopopliteal junction) (Fig 2B) (3,5,6).

Perforator Veins

Numerous small variable perforator veins normally function to contribute to the drainage of the superficial veins into the deep venous system. These veins are designated by using anatomic location (ankle, leg, knee, or thigh) combined with positional information (anterior, medial, posterior, or lateral) when the anatomic label is not locationally specific. The medial perforators of the thigh and leg are the most relevant, as they typically drain the GSV into the deep veins. The medial perforators of the thigh include the inguinal perforator vein and femoral canal perforator vein, both of which are defined by their position craniocaudal in the medial thigh. The medial perforators of the calf include the paratibial and posterior tibial veins. In the absence of disease, perforator veins are often small and cannot be easily detected at US without careful scrutiny (3).

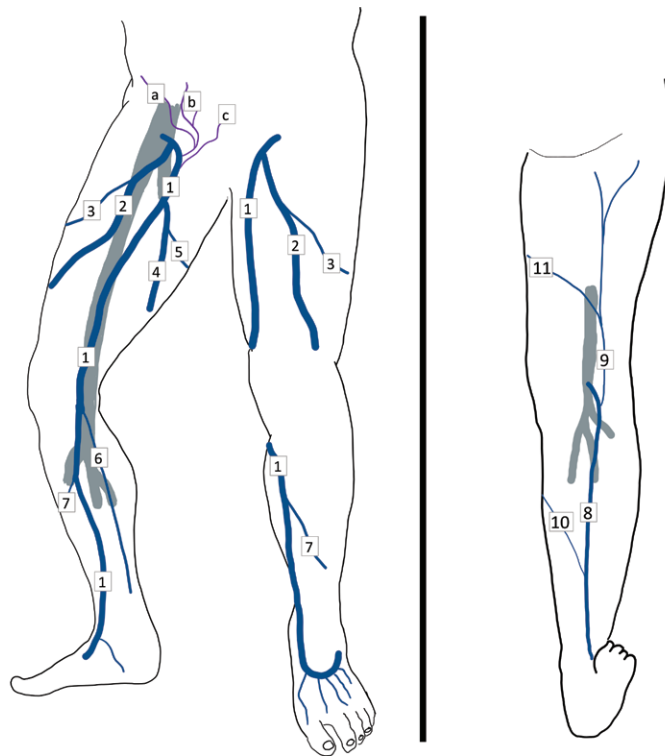


Figure 3. Schematic images of the superficial venous anatomy of the lower extremities. Superficial veins are labeled with numbers: GSV (1), AAGSV (2), anterior thigh circumflex vein (3), posterior accessory GSV (4), posterior thigh circumflex vein (5), posterior calf circumflex vein (6), anterior calf circumflex vein (7), SSV (8), cranial extension (9), intersaphenous vein (10), and posterior thigh circumflex vein (11). Connections to the abdominal and pelvic veins are labeled with letters: superficial circumflex iliac vein (a), superficial inferior epigastric vein (b), superficial external pudendal vein (c). Part of the deep veins of the right lower extremity are not labeled but are depicted underneath in gray.

Pathophysiology of Varicose Veins

Normally, venous sufficiency is maintained by a series of valves with two cusps that act as mechanical gates, allowing forward flow directed centrally but precluding flow reversal. In the deep venous system, these valves are not present in the iliac veins, with typically only one valve in the CFV (not always present), two to four valves in the FV, and approximately a dozen valves in the deep calf veins (7). In the superficial venous system, the GSV has a terminal valve at the top, with two to three additional valves in the thigh. The GSV and SSV have about 10–12 valves in the calf (7).

There are four anatomic pathways leading to lower extremity varicosities (8), and each pathway reflects an underlying pathophysiology in which increased venous pressure and valve dysfunction work in tandem to negatively affect each other, eventually disrupting the normal network of smooth-muscle cells and elastin fibers surrounding veins and causing reactive collagen deposition that leads to fibrosis and scarring, further impairing venous function (2,9).

Deep Venous Insufficiency with Perforator Vein Insufficiency

The incipient cause of deep venous insufficiency may vary between loss of valve function, which can occur as a consequence of deep vein thrombosis (DVT), and increased venous pressure, which can occur with congestive heart failure, but it ultimately results in valve dysfunction and increased

deep venous pressure working in tandem to exert retrograde force on perforator veins. These perforator veins normally function to drain superficial veins to the deep system, but the direction of venous flow reverses when these veins become insufficient, resulting in transmission of the deep venous increased pressure to the connected superficial vein segment. Increased pressure in the connected venous segment facilitates superficial vein valve failures along the length of the vein as each segment is sequentially affected (8).

Saphenofemoral and Saphenopopliteal Insufficiency

Varicose veins and venous insufficiency can also result from terminal valvular dysfunction at the connection where a major superficial vein drains into a deep vein. The saphenofemoral junction refers to the point at which the GSV drains into the CFV; the saphenopopliteal junction refers to the point at which the SSV drains into the PV (3). The terminal vein at either location can be damaged by constantly elevated venous pressure against the terminal valve, whereby the valve loses function over time, or by fibrosis that results from venous thrombosis involving the valve, whereby the valve does not return to normal function after vein recanalization (8). Once the terminal valve no longer functions, the vein is subject to reflux force that causes dilatation and upstream communication of increased pressure on the next valve, leading to a domino effect that causes multiple valve failures.

Calf Muscle Pump Causes

Many veins of the leg are contained within muscular compartments or within fibroelastic sheaths tethered to muscular compartments of the leg. When calf muscles contract, the tethering and compartmentalization compress veins, pumping venous blood forward (8). With calf muscle pump weakness, loss of this normal mechanical action in calf veins reduces the forward flow of blood, as can occur with obesity or leg immobility, resulting in venous stasis. Over time, stasis leads to dilatation and valvular incompetence (2). Alternatively and less commonly, perforator valves are subject to considerable strain when calf muscles with increased blood volume due to activity-related vasodilatation experience a sudden increase in pressure, as can occur with repetitive squats with weights. In patients who are predisposed to valve dysfunction, stress from this increased pressure can precipitate valve failure over time.

Pudendal Vein and Pelvic Vein Insufficiency

Pudendal and pelvic veins have small tributaries that connect with the GSV and superficial venous system of the lower extremity. The most common of these communicating veins are the superficial external pudendal vein, superficial epigastric vein, and superficial circumflex iliac vein. Pudendal vein and pelvic vein insufficiency exerts hydrostatic force on the superficial veins through these connections in the same way that deep venous insufficiency does through perforator veins. These connections reflux into the GSV, and the subsequent venous stasis and increased pressure within the GSV lead to varicosities (8).

Clinical Assessment and Treatment Methods

In the 2004 update of the Clinical-Etiology-Anatomy-Pathophysiology (CEAP) clinical classification system for chronic venous disorders, seven clinical manifestations, four etiologies, four anatomic distributions, and four basic pathophysiologic mechanisms are categorized (10). This article is focused on the venous US evaluation of patients who have varicose veins (patients in CEAP class C₂) with superficial anatomy manifestations. For varicose veins, the primary purpose of the sonographic evaluation is not to document where the varicosities exist but rather to understand the anatomy and physiologic features that cause the varicosities, as well as those that potentially complicate treatment. The treating physician's approach to addressing varicosities in any given patient will depend on a number of factors, including treatment goals.

While optimal treatment eradicates the pathophysiologic cause of the varicosities, palliative treatment may be pursued even if the pathophysiologic cause cannot be cured. For example, when there is reflux in the GSV at the saphenofemoral junction without coexisting deep venous insufficiency, there is the chance that ablation of the entire GSV with local treatment of existing varicosities will address the underlying cause (superficial venous hypertension from saphenofemoral reflux related to terminal valve dysfunction), with a presumed lower risk of recurrence. If there is GSV insufficiency with coexisting deep venous insufficiency, then ablation of the GSV will almost certainly not be curative because the reversed flow in perforating veins will not be addressed. However, the ablation may still have palliative value if deep venous insufficiency can be controlled with compression hose. US findings that have potential implications for treatment are outlined in the Table.

While a comprehensive discussion of varicose vein treatment is beyond the scope of this article, sonographers and sonologists can benefit when they are familiar with the variety of methods to treat varicose veins, since the treatment choice is based on imaging observations. Methods vary in invasiveness and best application, with endovenous laser ablation (ELA), sclerotherapy, ambulatory phlebectomy, and vein stripping being the most common options.

Minimally Invasive Treatment

ELA involves the use of an intravenous laser fiber to apply energy that heats the vein, usually the GSV, causing long-term obliteration by damaging the endothelial lining and leading to fibrosis and scarring. Most health care professionals use the term *endovascular heat-induced thrombosis* to refer to this process. Endovascular heat-induced thrombosis is generally the preferred method and first-line treatment for varicose veins, when applicable (11,12). To perform ELA, the operator uses a catheter to insert a laser fiber into the affected vein at a peripheral location and advance the tip to near its draining point. For the GSV, the catheter is usually inserted into the upper calf, and the tip of the laser fiber is subsequently advanced to near the saphenofemoral junction, peripheral to the insertion of the inferior epigastric vein so that preserved flow in the inferior epigastric vein will assist in preventing the extension of thrombosis into the CFV. Once in position, the laser is activated and gradually pulled back through the vein. Doppler US evaluation performed shortly after the procedure serves to document thrombosis of the treated vein and ensure that the thrombus does not extend into the CFV. The occluded GSV narrows as fibrosis ensues, with long-term

Potential Implications of Specific Venous US Patterns	
Observation or Pattern	Potential Implications
Saphenofemoral insufficiency with GSV reflux in mid thigh	GSV ablation may address the pathophysiologic cause Ensure the absence of a coexisting dilated accessory GSV, which may need to be separately addressed
Saphenofemoral insufficiency without GSV reflux in mid thigh	Ensure the absence of an insufficient accessory GSV (AAGSV or PAGSV) that carries the reversed flow to varicosities, since the accessory vein will need to be addressed
Varicosities around saphenofemoral junction without saphenofemoral insufficiency	Consider a pudendal vein or pelvic vein source of the varicosities Consider central venous obstruction
Upper-thigh GSV reflux without saphenofemoral insufficiency	Ensure the absence of an insufficient pudendal vein or pelvic vein connecting to the upper GSV If there is a coexisting deep vein insufficiency, the GSV reflux is probably due to perforator vein(s)
Lower-thigh GSV reflux without saphenofemoral insufficiency	Ensure there is no connection to an insufficient SSV through cranial extension and a posterior circumflex vein in the thigh or calf If there is a coexisting deep vein insufficiency, the GSV reflux is probably due to perforator vein(s)
No saphenofemoral or saphenopopliteal insufficiency	If there is a coexisting deep vein insufficiency, the varicosities are probably due to perforator vein(s) If there is no deep vein insufficiency, the varicosities may reflect a calf muscle pump issue
GSV superficializes	Ablation may not be possible
GSV more than 8–10 mm in diameter	Ablation may be ineffective
Small GSV	The GSV is unlikely to be the source of clinical concerns

Note.—PAGSV = posterior accessory GSV.

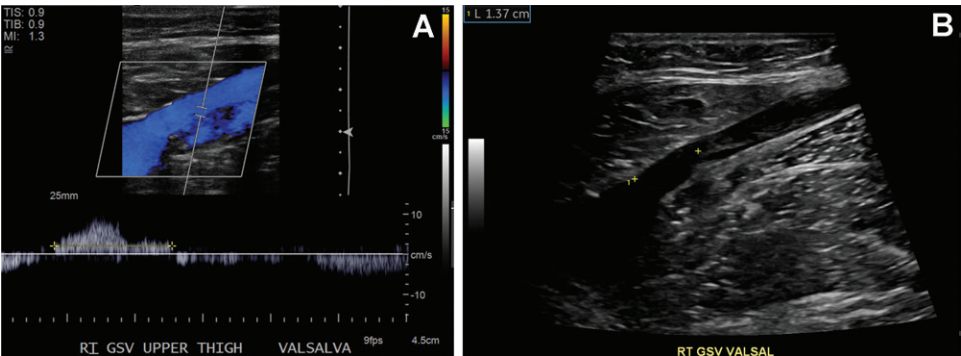


Figure 4. Right lower extremity venous insufficiency in a 74-year-old man before and after ELA. (A) Duplex Doppler US image before ELA shows reversed flow for 1.8 seconds in the proximal GSV with the Valsalva maneuver. (B) Longitudinal US image of the GSV after ELA shows thrombosis extending 1.3 cm distal to the saphenofemoral junction.

occlusion rates reported as high as 100% at 3 years (13). Others have shown a more realistic Doppler US–confirmed failure rate of 30% at 6 years, with a 12% clinical failure rate in the same group (Fig 4) (14). Higher rates of failure occur when the vein is 8 mm or larger because the laser energy may fail to adequately affect both walls (14); tumescent anesthesia can be used to try to contract the vein for improved outcomes. ELA can cause skin burns if the treated vein is too close to the dermis or there is reflux of heated blood into a large varicosity too close to the dermis (15).

Sclerotherapy involves the direct injection of a sclerosing agent (osmotic agents, detergents, irritants, corrosives, or foam) to destroy the venous

endothelium. Damage to the endothelium leads to fibrosis and obliteration of the vein. Sclerotherapy is often used to treat small spider veins and telangiectasias, although foam sclerotherapy is an alternative approach to addressing larger superficial veins when ELA is contraindicated (16). Ambulatory phlebectomy can be used to treat larger superficial varicosities that are visible at physical examination. A small skin incision is made to expose the varicosity, small hook instruments pull the varicosity through the skin incision, and the varicosity is ligated and excised. This technique requires skin incisions at the site of each varicosity being treated. While this procedure can be used to treat individual varicosities, it does not address the underlying cause (17).

Invasive Treatment

Vein stripping is the most invasive method of treating GSV insufficiency, and compared with minimally invasive strategies, it is associated with higher morbidity (18). This approach is generally reserved for scenarios in which minimally invasive methods are not appropriate. The vein-stripping technique is variable and partially dependent on the specific stripping device used. In general, the target vein is dissected at its upper end and the upper branches are divided. The vein is then accessed and divided from other branches. A wire and/or stripping instrument is passed through the target vein from the lower end to the divided upper end, and the instrument is secured at the upper and lower ends. The secured device, and the vein along with it, is then removed from the patient (18–20).

Doppler US Protocols

Patient Positioning

A complete diagnostic evaluation for varicose veins involves a full examination of the deep veins and the superficial veins from the inguinal ligament to the ankle and relies on color and spectral Doppler US. Whereas lower extremity duplex Doppler US examinations to look for DVT can be performed with the patient in the supine or semi-Fowler position, this is not acceptable when the goal of the examination is to evaluate for chronic venous insufficiency. For insufficiency studies, the legs must be positioned below the level of the patient's head to maximize venous filling and optimize evaluation for reflux (3); this is optimally achieved by having the patient lie on a tilt table in the reverse Trendelenburg position at a 60° or greater incline. Evaluation of patients in the standing position is theoretically ideal but practically difficult for both the patient and the sonographer.

The leg being examined should not be weight bearing, with the hip externally rotated and the knee slightly flexed (Fig 2A) (21). This position provides exposure of the deep veins from the medial thigh to the calf without patient repositioning (3). Right and left lateral decubitus positions with the patient tilted may be needed to evaluate the SSV. The left lateral decubitus position is appropriate for evaluation of the right SSV, and the right lateral decubitus position is appropriate for evaluation of the left SSV (Fig 2B) (3,21).

Performing the Examination and Documenting Examination Results

The methodology and best practices for evaluation of DVT have been well described (21). Briefly, evaluation consists of compression with gray-scale still or cine US documentation of the

CFV, saphenofemoral junction, deep FV, and FV at the upper thigh; FV at the mid thigh; FV at the distal thigh; PV; posterior tibial vein; and peroneal veins. If any of the deep veins are non-compressible, additional images may be needed to clearly document the extent of the abnormality (21). According to practice parameters, long-axis color and spectral Doppler US images should be obtained at both the CFV and the PV for the evaluated leg, with additional long-axis Doppler US images obtained in abnormal areas. When reviewing the images obtained by a sonographer, the interpreting physician should pay attention to ensure that the compression documentation is adequate, reviewing the long-axis Doppler US images for spectral waveform clues of downstream venous obstruction and the color Doppler US images for filling defects and wall thickening.

Evaluation of the lower extremities for venous insufficiency, in contrast to DVT detection, requires more time and scrutiny to ensure that the cause of the insufficiency is properly understood (7,22). The 2019 revised ACR-AIUM-SPR-SRU (American College of Radiology, American Institute of Ultrasound in Medicine, Society for Pediatric Radiology, Society of Radiologists in Ultrasound) practice parameter relevant to venous insufficiency is less prescriptive regarding the specific images and documentation needed for the evaluation, indicating that duplex US interrogation should be performed at “as many levels as necessary to ensure a complete examination based on the clinical indications and a standard protocol” (21).

Color and spectral Doppler US examinations play a much larger role in assessments for venous insufficiency than in those for DVT evaluation, and provocative maneuvers are required, as spontaneous reflux is uncommon. The Valsalva maneuver is an appropriate provocative maneuver for the saphenofemoral junction, while augmentation with use of calf compression (applied manually or by using a rapid cuff inflation system) is necessary for provocative maneuvers beyond the saphenofemoral junction when the terminal valve at the saphenofemoral junction is intact (3,21).

We recommend that the sonographer ask the patient to point out symptomatic areas before images are obtained and briefly reflect on the distribution of visible varicosities in relation to the expected course of the GSV, AAGSV, and SSV (Fig 3). Doing this helps ensure that atypical patterns are recognized. Standards promulgated by the Society of Vascular Ultrasound stipulate that at minimum, baseline and maneuver response spectral Doppler US documentation be obtained at

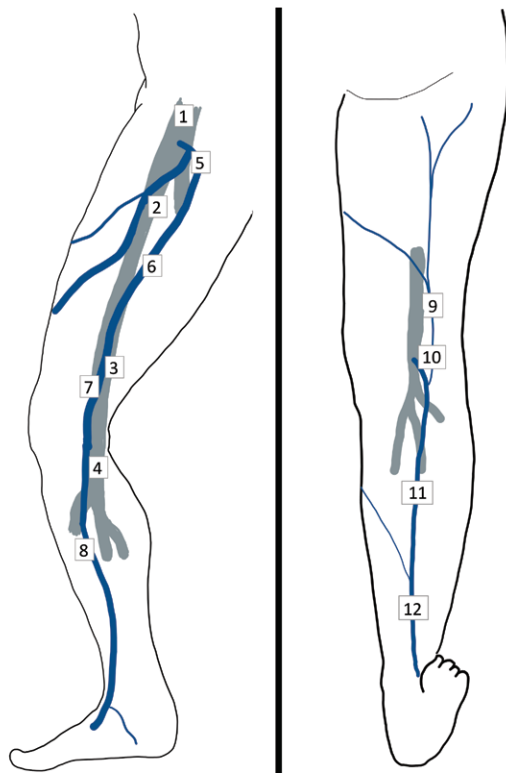


Figure 5. Drawings of the right leg show the location of protocol images for venous insufficiency evaluation (after compression images are obtained for evaluation of DVT) at one institution. Left: For evaluation of the deep veins and GSV, longitudinal duplex color and spectral Doppler US images of the CFV (1), upper FV (2), lower FV (3), and PV (4) are obtained by using provocative maneuvers. Subsequently, longitudinal duplex Doppler and transverse gray-scale US images of the GSV at the saphenofemoral junction (5), mid thigh (6), lower thigh (7), and calf (8) are obtained. The anteroposterior diameter of the GSV is measured at locations 5–8. Right: For evaluation of the SSV, longitudinal duplex Doppler and transverse gray-scale US images of the cranial extension of the SSV (if present) (9), and at the saphenopopliteal junction (10) and mid calf (11) are obtained. Distal-calf SSV imaging is included when stasis ulcers are present. Transverse cine sweep images of the GSV, AAGSV, and SSV are acquired and usually suffice to show varicosities, but additional cine sweep images of unusual or large clustered varices are added as needed. In addition, images of abnormal findings such as enlarged perforators are obtained when these are encountered.

the CFV, mid FV, PV, GSV at the saphenofemoral junction, GSV above and below the knee, SSV at the saphenopopliteal junction, and SSV in the calf (Fig 5) (22). At our institution, gray-scale US images of the GSV and SSV at these locations are also captured in the transverse plane to ensure that the vein lies within the expected fascial envelope, understand the vein margin distance from the skin surface, and measure the anteroposterior diameter of the vein.

In the thigh and upper calf, the GSV is usually found in a well-formed fascial envelope called the “saphenous eye” or “Egyptian eye” (Fig 6). This is an important finding to recognize because escape of the vein from the envelope to a superficial location, a condition that we

term *superficialization*, will influence treatment options. While the Valsalva maneuver is sufficient to test for reflux in the CFV and upper FV and at the saphenofemoral junction, normalcy of valve function at these levels requires a switch to an augmentation maneuver below. When reflux is present, the reflux time is measured from the start of flow reversal within the vein after provocative maneuvers are initiated to the time that the reversal of flow ceases (Fig 6). Normal valves do not close instantaneously, and brief reflux is considered normal (3).

Throughout the entire examination, variant anatomy and important accessory veins—the AAGSV (Fig 7) and cranial extension of the SSV (Fig 8) in particular—merit scrutiny (21). While the saphenofemoral junction is often the most cephalic point of the superficial venous system that is typically evaluated, patients with variant anatomy may have superficial veins that extend above the saphenofemoral junction and may require interrogation and documentation for adequate understanding and documentation of the anatomic pathway for varicosities when these are relevant (3,5).

Because the superficial veins have substantial anatomic variability, cine clip imaging and documentation in the transverse plane from just above the saphenofemoral junction down to the knee are invaluable. In our practice, two cine clip sweep images of the GSV are obtained: one centered on the GSV along the entire thigh to the upper calf (Movie 2) and the other centered on the AAGSV from the saphenofemoral junction to the point at which it diminishes in size in the subcutaneous tissues (Movie 1). An additional transverse cine clip sweep image along the course of the SSV can be helpful for showing anatomic variations or clarifying abnormal observations in select cases (Movie 3). These cine clips can help to document the connection of any varicosities to the GSV, AAGSV, and SSV, as well as depict dilated perforator vein connections to these vessels. They also highlight abrupt changes in GSV or SSV caliber, which can indicate a source of reversed flow, such as at the site of the perforator connection. When varicose and perforator connections have been adequately captured on these cine clip sweep images, the interpreting physician is better able to communicate relevant observations that may influence treatment. Of course, sonographers should routinely perform real-time imaging to ensure that there are no other confounding anatomic features that need image capture to accurately convey the condition.

Reviewing and Reporting Examination Results

Radiologists and other interpreters of vascular US studies benefit from understanding the

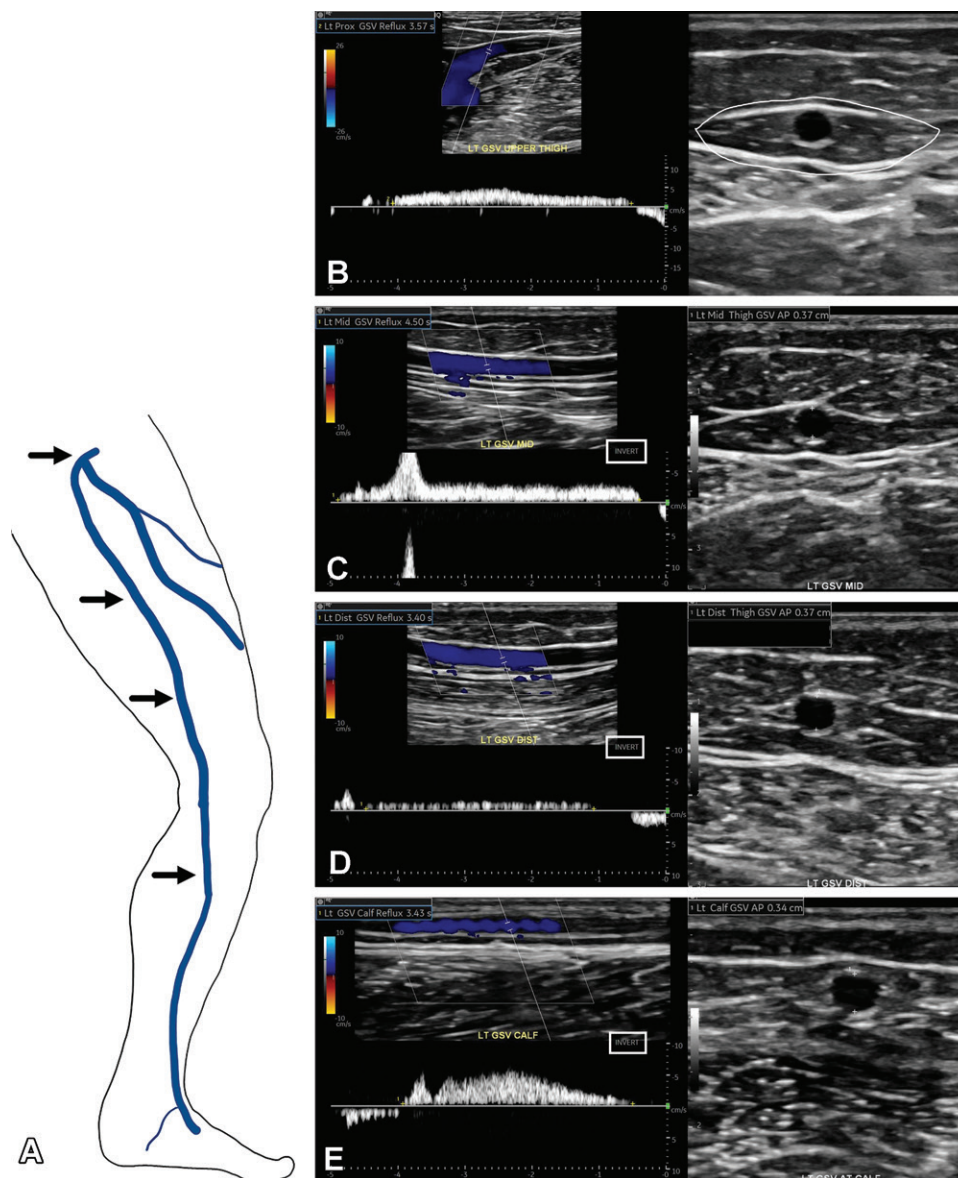


Figure 6. GSV insufficiency through the entire leg of a 23-year-old man. (A) Drawing of the left leg and the course of the GSV shows the areas examined with US (arrows). (B–E) Spectral Doppler US images obtained with the Valsalva maneuver (left) and transverse US images (right) at the saphenofemoral junction (B), mid thigh (C), distal thigh (D), and upper calf (E) show venous insufficiency with reversed flow within the vein, which lies within the saphenous fascial envelope (ie, saphenous eye). In B (right image), the boundaries of the saphenous fascial envelope are outlined. The white boxes (*INVERT*) in C, D, and E indicate that the spectral Doppler waveform display is inverted, accounting for the spectral signal of flow moving away from the transducer that is depicted above the baseline. The color signal shown on the color Doppler display often does not represent the predominant direction of flow during a provocative maneuver.

technical aspects of duplex Doppler US and the challenges faced by sonographers in obtaining images that convey accurate diagnostic information, to avoid interpretation error and/or confusion. For example, the color Doppler anatomic portion of a duplex Doppler US image is not always representative of the captured spectral Doppler timeline. The color Doppler signal represents one moment in time that may not represent the predominant direction of flow over time or with maneuvers. The spectral display represents the direction and velocity of flow

as a function of time. Careful attention to the spectral Doppler scale enables the reviewer to recognize when the display has been flipped such that flow with the vector toward the transducer, which has a positive value, is displayed below the spectral baseline, and vice versa. Some manufacturers highlight this condition by inserting the word “invert” near the spectral Doppler scale (Fig 6C–6E).

Detailed reporting of an examination for venous insufficiency ensures that a patient is well suited for treatment, influences insurance reim-

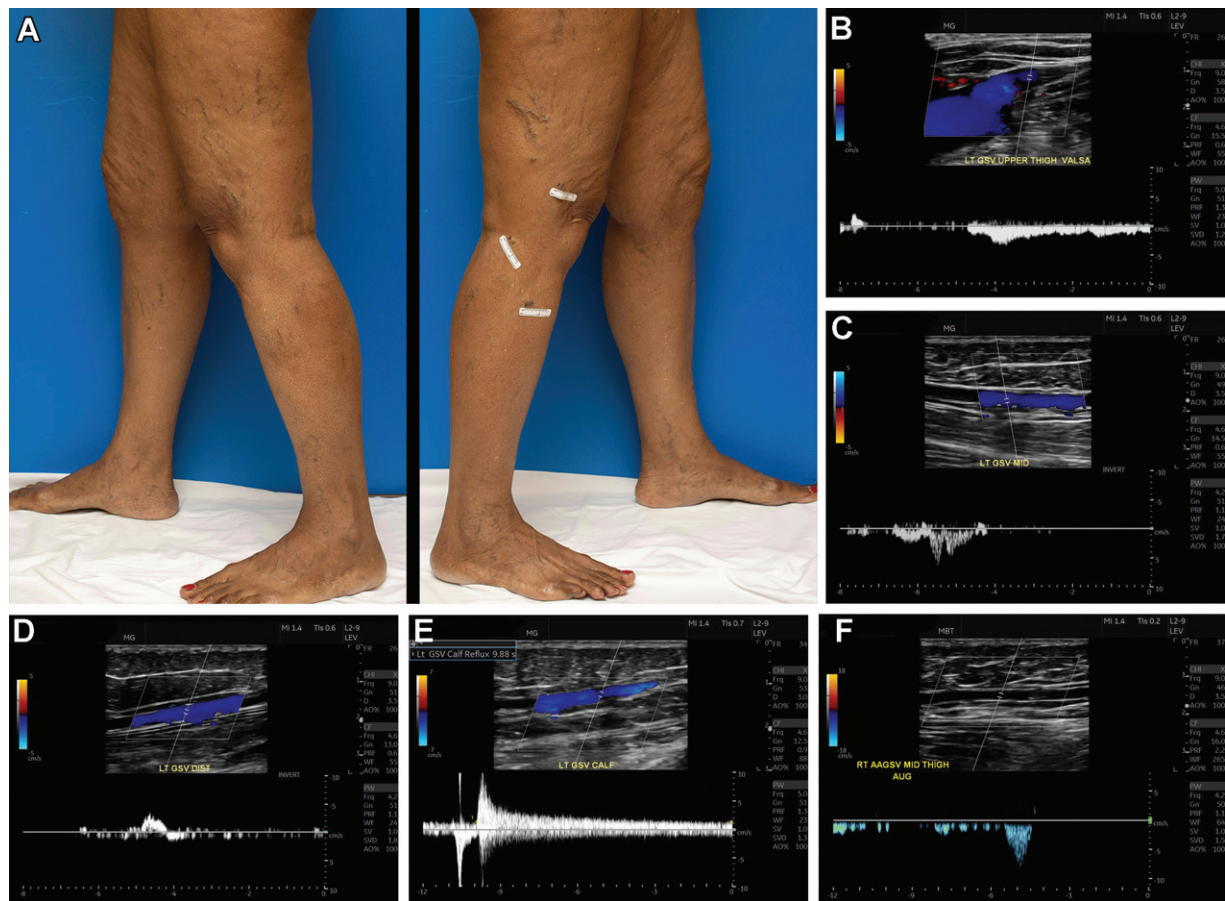


Figure 7. Bilateral lower extremity discomfort over engorged spider telangiectasias and palpable varicosities, with no relief despite the use of compression hose for 1 year, in a 60-year-old woman. (A) Side photographs of both lower extremities show that the lateral zones of the lower thighs and the legs are the most affected by telangiectasias. At clinical examination, the most symptomatic varicosities were in the right lower extremity (marked with rulers). (B–F) Spectral Doppler US images show no saphenofemoral insufficiency bilaterally and no reflux along the course of the right GSV. However, prolonged reflux in the left GSV at the calf is depicted. The right AAGSV and associated branches were scrutinized (F), given the predominantly right-sided symptoms, and were normal at Doppler US evaluation (Movie 1). The vascular surgeon performed sclerotherapy of the palpable right lateral varicosities and associated telangiectasias. Despite the presence of left-calf GSV reflux at Doppler US evaluation, left GSV treatment was not performed.

bursement, and guides appropriate treatment. Templated reporting helps ensure that all key points in the examination are properly communicated. We find it useful to have the examining sonographer include a diagram to briefly note where deep or superficial veins are insufficient. We believe that having sonographers write out the actual measurements taken during the examination is tedious and wastes time, and we prefer to use DICOM (Digital Imaging and Communications in Medicine) structured results data to automatically fill these measurements into the report.

When venous insufficiency is demonstrated to last longer than 0.5 second, the most important parameters to report are the location where reflux was observed and the reflux time. Although specification of the maneuver that elicited the reversed flow (Valsalva vs augmentation) adds context, this detail may not be critical. A reflux time longer than 500 msec, or 0.5 second, is the

conventional threshold for insufficiency in the deep and superficial venous system, but some advocate 1 second as a better threshold to improve specificity (1). Labeling instances in which deep venous reflux was demonstrated to last more than 0.5 second but less than 1.0 second as “equivocal” rather than simply concluding that these indicate an abnormality may better reflect the trade-off between sensitivity and specificity.

As discussed earlier, deep venous insufficiency causes superficial varicosities by way of reversed flow in perforator veins. Describing the location of dilated perforator veins with reversed flow helps to explain the anatomy of the patient’s problem, even if doing so does not typically change management. Detailed recording and reporting of the size of the deep veins is not important because it has no effect on the treatment. On the contrary, the sizes of the GSV and SSV at the specified locations help the treating physician

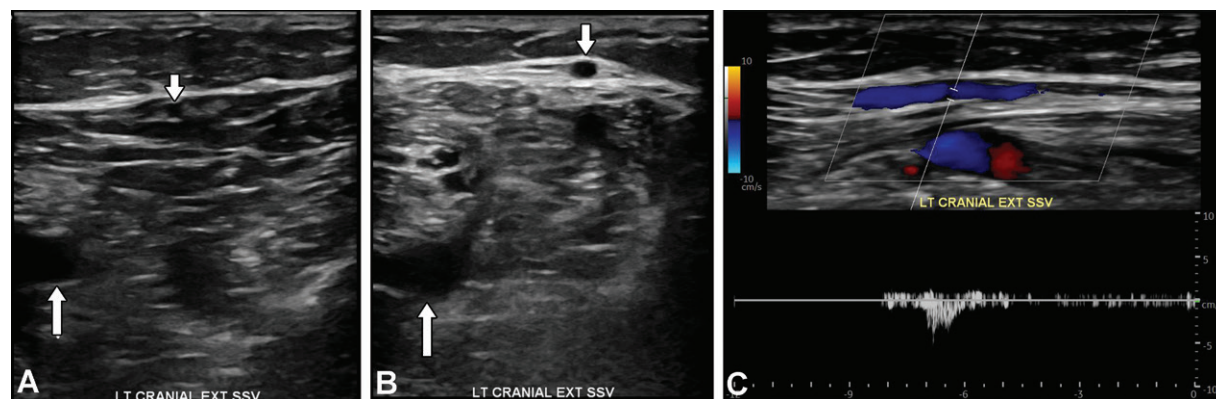


Figure 8. Cranial extension of the SSV, also known as the vein of Giacomini, with no saphenopopliteal junction, in a 60-year-old woman. (A) Transverse US image of the cranial extension of the SSV (short arrow) obtained with the transducer on the posterior thigh shows the vein remaining in the fascial envelope at the level of the junction of the PV with the FV (long arrow). (B) Transverse US image shows the SSV (short arrow) at the mid calf. The long arrow points to the main gastrocnemius vein. (C) Longitudinal duplex Doppler US image shows an absence of reflux in the cranial extension of the SSV.

recognize when there may be difficulties with targeting or access (when too small) or laser ablation (when too large) (Fig 9). Highlighting the time at which the GSV is superficial to the fascial envelope is critical, because laser ablation of the GSV too close to the skin surface subjects the patient to an increased risk of skin burn (Fig 10) (Movie 4) (21).

The presence of SSV cranial extension (ie, the vein of Giacomini) is common, but highlighting the time at which cranial extension has substantial reflux focuses attention on a potential confounding pathophysiology. For instance, the observation of cranial extension with substantial reflux can increase suspicion for intersaphenous communication, accounting for varicosities in the posterior compartment arising from GSV insufficiency (5).

While detailed narrative descriptions have no substantial benefit in cases of straightforward patterns of venous insufficiency beyond the listing of reflux locations and times, and vein diameters, detailed narratives do help communicate important features when more complicated patterns and connections exist. A suggested report template (Appendix E1) and example report (Appendix E2) are provided as supplemental material.

Patterns and Pitfalls

Saphenofemoral or saphenopopliteal insufficiency related to terminal valve dysfunction in the GSV or SSV leads to straightforward patterns of Doppler US abnormalities that are easy to understand and report (Fig 6). However, many cases are not so straightforward; recognizing interpretation pitfalls helps ensure that a miscommunication to the clinician caring for the patient does not lead to suboptimal treatment strategies. Understanding

major potential pitfalls also helps sonographers and sonologists recognize and detect important anatomic and pathophysiologic details when imaging findings are confusing (Table).

Pitfall: Not Recognizing or Detecting Deep Venous Reflux

Puggioni et al (23) reported failed superficial vein ablation in approximately two-thirds of patients with underlying deep venous reflux. If deep venous reflux is not recognized, the treatment for superficial venous reflux may be ineffective (Fig 11). Deep venous duplication is an important anatomic variation to scrutinize, as insufficiency may involve only one of the duplicated veins (Fig 12).

Pitfall: Misidentification of CFV Reflux as Deep Venous Reflux When Isolated or Related to Saphenofemoral Insufficiency

Mild reflux in the CFV can be normal when it is not associated with flow into another branch, reflecting the variable presence of a valve in the CFV (Fig 13). Furthermore, when the terminal valve of the GSV is insufficient, reflux will be demonstrable in the short segment of the CFV, leading to the next most proximal valve in the CFV (Fig 14). Reflux limited to this segment of the CFV should not be mislabeled as deep venous insufficiency, or a narrative context in the report is required to ensure that this reflux is not misconstrued as true deep venous reflux, potentially causing inappropriate patient exclusion from laser vein treatment (24).

Pitfall: Not Recognizing AAGSV Involvement in Saphenofemoral Insufficiency

The observation of saphenofemoral insufficiency usually leads to the use of ELA. Treatment



Figure 9. Saphenofemoral insufficiency in a 44-year-old woman. (A) Anterior (left) and medial (right) clinical photographs show obvious varicosities in the distribution of the right GSV. (B) Longitudinal duplex Doppler (left) and transverse (right) US images of the GSV at the saphenofemoral junction show prolonged reflux into the dilated GSV, lasting more than 3 seconds (not measured on this image), with the GSV measuring 10 mm. The patient has a higher risk of laser vein ablation failure owing to the size of the GSV; this is important to communicate to the treating clinician.

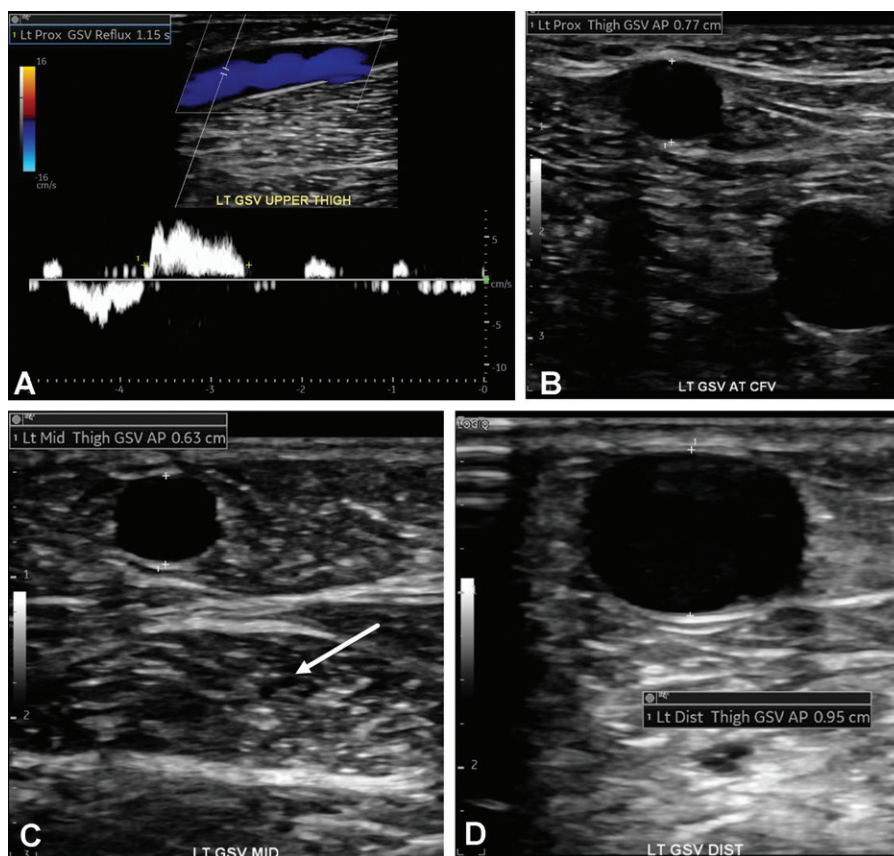
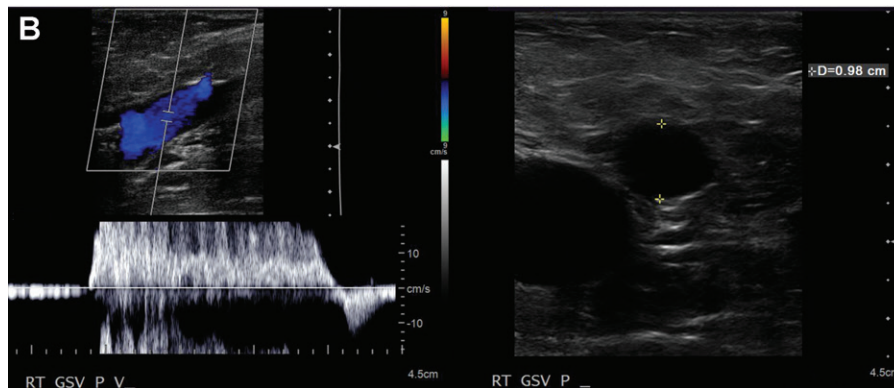


Figure 10. Left venous insufficiency in a 72-year-old woman. (A) Duplex Doppler US image of the left upper GSV shows saphenofemoral insufficiency, after the Valsalva maneuver, lasting 1.2 seconds. (B) Findings on the transverse sonogram of the GSV in the upper thigh confirm the presence of the vein in the fascial envelope. (C) Transverse sonogram of the GSV in the mid thigh shows that the vein has superficialized and is now located above the fascial envelope. Arrow points to the typical location of the GSV within the fascial envelope. (D) Transverse sonogram of the GSV at the distal thigh shows the GSV immediately under the skin. Moreover, the vein is now 9.5 mm in diameter. The patient is at increased risk for complications and treatment failure if laser vein ablation is attempted.

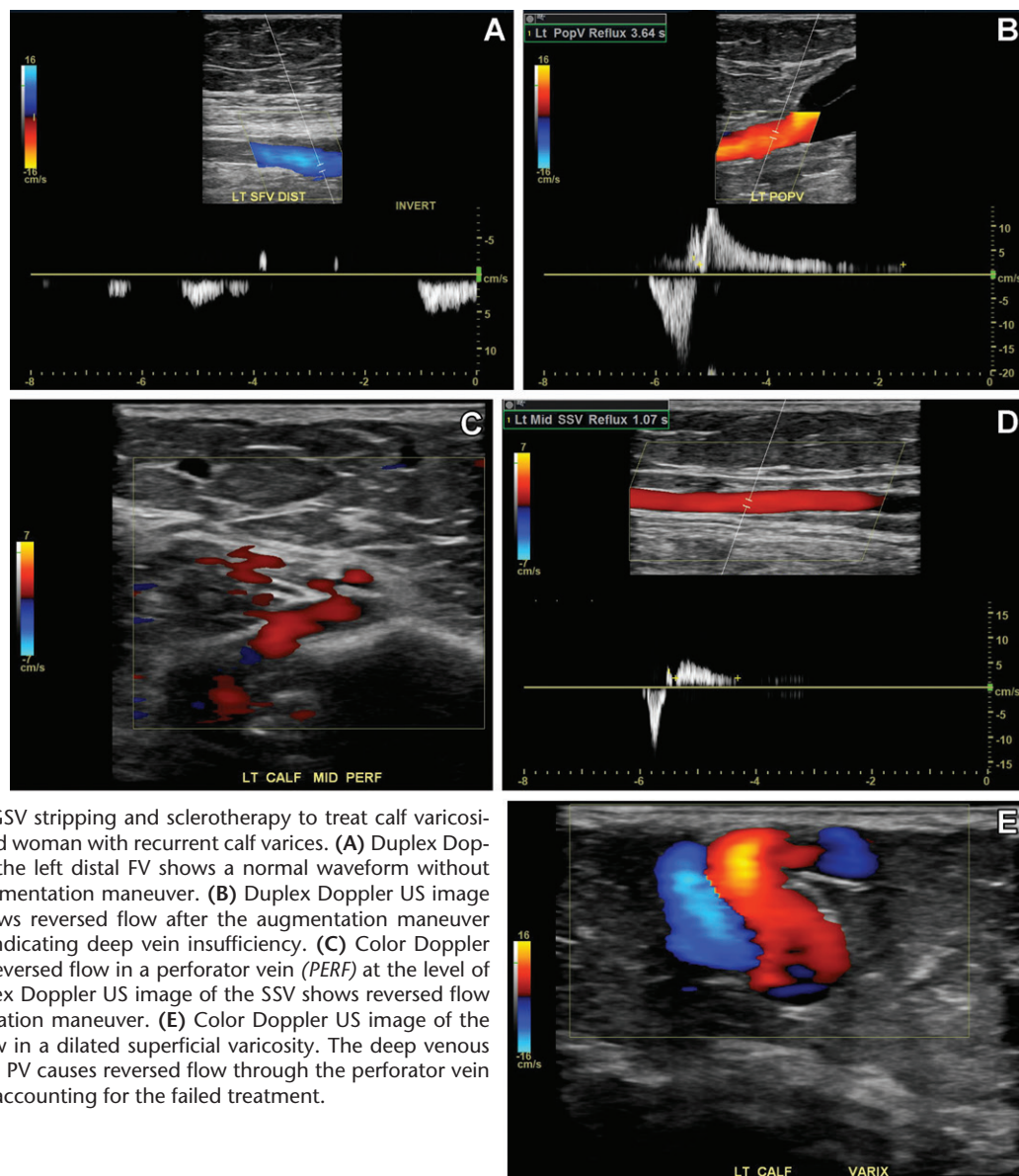


Figure 11. Left GSV stripping and sclerotherapy to treat calf varicosities in a 63-year-old woman with recurrent calf varices. (A) Duplex Doppler US image of the left distal FV shows a normal waveform without reflux after an augmentation maneuver. (B) Duplex Doppler US image of the left PV shows reversed flow after the augmentation maneuver for 3.6 seconds, indicating deep vein insufficiency. (C) Color Doppler US image shows reversed flow in a perforator vein (PERF) at the level of the calf. (D) Duplex Doppler US image of the SSV shows reversed flow with the augmentation maneuver. (E) Color Doppler US image of the left calf shows flow in a dilated superficial varicosity. The deep venous insufficiency at the PV causes reversed flow through the perforator vein and into the SSV, accounting for the failed treatment.

failures occur when one does not recognize that the reflux involves the AAGSV instead of or in addition to the GSV (Fig 15) (25). If the reflux involves the AAGSV only, then laser vein ablation of the GSV is inappropriate, because the laser should have been inserted into the peripheral portion of the AAGSV (Movie 5). The AAGSV may be amenable to laser ablation when more than 10 cm of the vein is straight. If the reflux involves both the GSV and the AAGSV, ablation of the GSV alone will lead to an incomplete response.

Pitfall: Not Recognizing or Reporting GSV or AAGSV Superficialization

If an insufficient GSV or accessory GSV leaves the fascial envelope and courses near the dermis, there is a risk of the ELA burning the skin. Not recognizing or reporting this situation frustrates the operator performing the procedure, who may recognize this situation during the proce-

dures when using US to access the vein or when advancing the laser into the vein, leading to a last-minute procedural cancellation (Fig 10).

Pitfall: Central Venous Obstruction

Central venous obstruction can contribute to lower extremity venous insufficiency and potentially affects as many as 30% of patients with evidence of venous insufficiency (26). The lower extremity vein valves may become insufficient owing to venous dilatation in a patient with central flow restriction or obstruction, resulting in abnormal Doppler US findings with augmentation maneuvers and leading to peripheral treatment interventions. However, if the possibility of central obstruction is not recognized, there is the risk of these peripheral interventions being ineffective, as other dilated veins develop over time. Clues to possible central obstruction

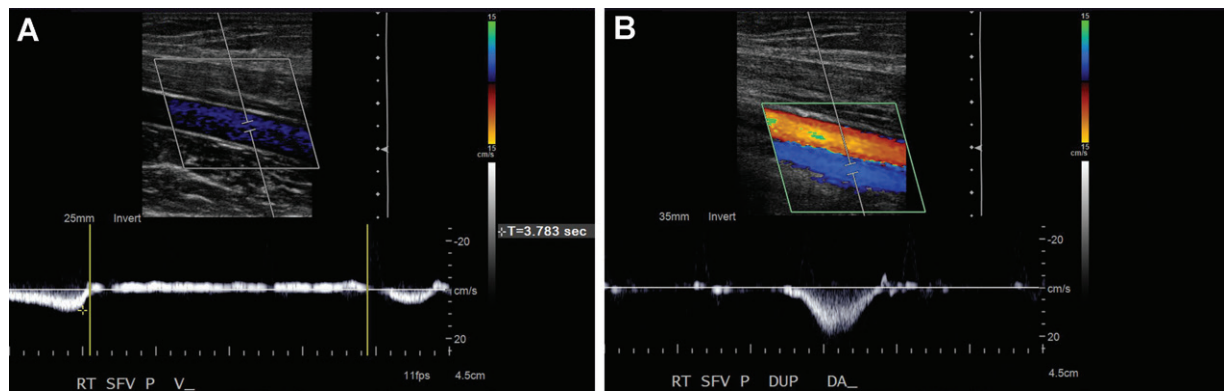


Figure 12. Duplicated FV, with deep venous reflux in only one of the veins, in a 47-year-old man. (A) Duplex Doppler US image of one of the FVs in the upper thigh shows extensive reflux (after an augmentation maneuver) that lasted 3.8 seconds. (B) Duplex Doppler US image of the adjacent duplicated FV shows no deep venous insufficiency after the augmentation maneuver.

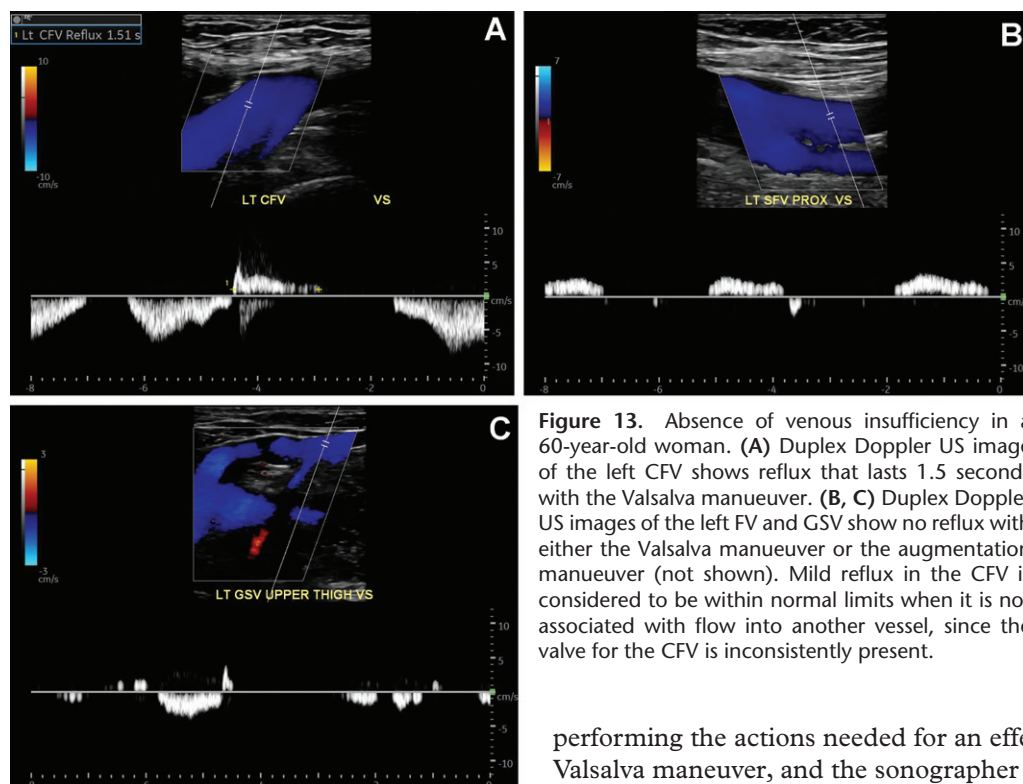


Figure 13. Absence of venous insufficiency in a 60-year-old woman. (A) Duplex Doppler US image of the left CFV shows reflux that lasts 1.5 seconds with the Valsalva maneuver. (B, C) Duplex Doppler US images of the left FV and GSV show no reflux with either the Valsalva maneuver or the augmentation maneuver (not shown). Mild reflux in the CFV is considered to be within normal limits when it is not associated with flow into another vessel, since the valve for the CFV is inconsistently present.

include loss of normal phasicity in the CFV and the detection of unusual collaterals such as dilated superficial epigastric veins, pudendal veins, or upper posterior thigh veins. When central obstruction is suspected, determining its location and extent typically requires venography or other cross-sectional imaging (Fig 16).

Pitfall: Ineffective Provocative Maneuvers

Assessment of venous insufficiency relies on the execution of effective provocative maneuvers that create a sufficiently high pressure gradient across the venous segment being examined, to maximize the demonstration of valvular dysfunction. Some patients have difficulty understanding or

performing the actions needed for an effective Valsalva maneuver, and the sonographer may be required to rely primarily on the augmentation maneuver (Fig 17). One method of overcoming this limitation is to instruct the patient to take a deep breath and resist pressure that is applied to the abdomen while holding his or her breath. The sonographer pushes on the patient's abdomen during the breath hold, and the patient's resistance to the push simulates the Valsalva maneuver. One can assess the adequacy of the Valsalva maneuver or simulated Valsalva maneuver by confirming substantial diminished venous return in the spectral waveform, which usually causes cessation of forward flow (Figs 7B, 11A). Similarly, effective augmentation can be documented as a spike of forward flow during the maneuver (Figs 12B). Identifying the expected waveform changes during either maneuver, even without

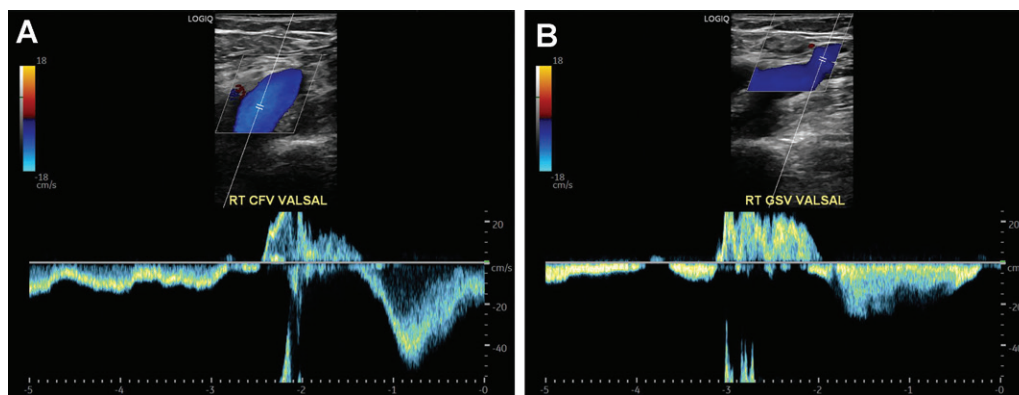


Figure 14. GSV insufficiency in a 66-year-old woman undergoing evaluation for DVT. (A) Duplex Doppler US image with a sample gate in the right CFV shows reflux. (B) Findings on Doppler US image with a sample gate in the GSV confirm that the distal right CFV is related to extension into the GSV.

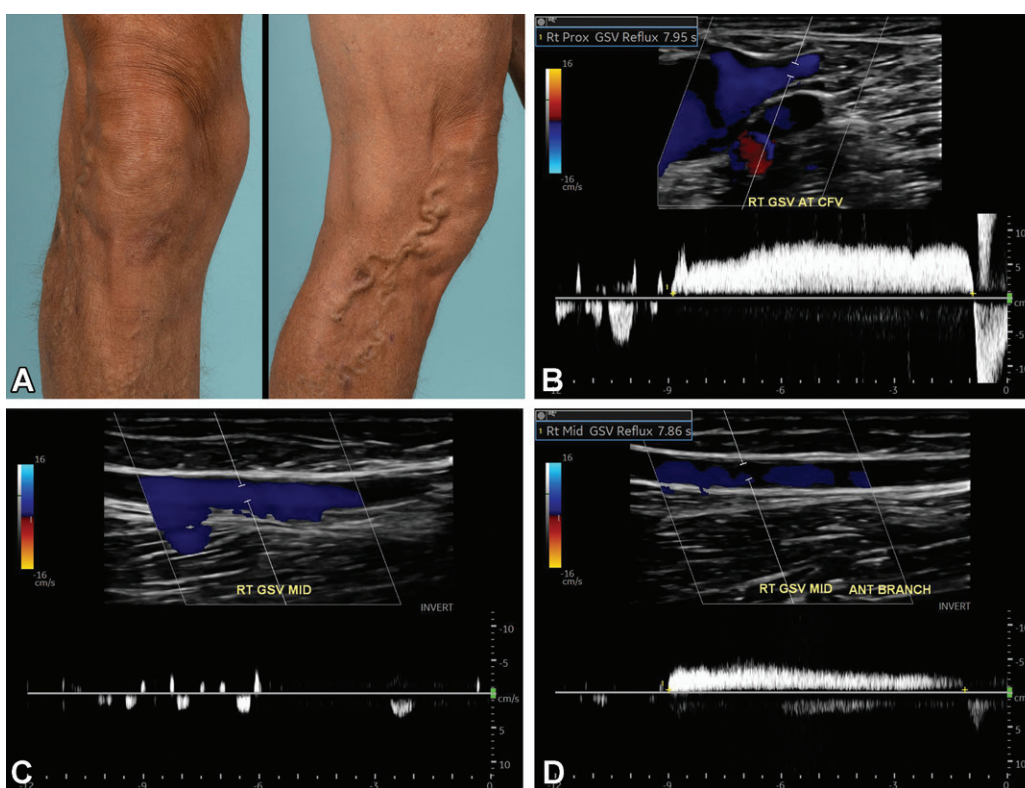


Figure 15. Right AAGSV insufficiency in a 62-year-old man. (A) Clinical photographs of the right leg show varicosities in the lateral aspect of the knee. This is the territory of the AAGSV, not the GSV. (B–D) Duplex Doppler US images of the saphenofemoral junction (B), GSV (C), and AAGSV (D) show that the insufficiency is exclusive to the AAGSV and does not extend down the GSV. Laser vein ablation would need to target the AAGSV, not the GSV.

observing any associated flow reversal, provides evidence of the proper maneuver execution.

Conclusion

Venous insufficiency causes significant morbidity. Successful Doppler US evaluations are based on detailed examinations of key locations of the superficial and deep venous system, with appropriate documentation and communication of relevant observations. Armed with an understanding of the lower extremity venous anatomy,

venous insufficiency pathophysiology, basic treatment strategies, protocol best practices, observation patterns, and diagnostic pitfalls, those who perform and interpret these studies are better prepared to serve as effective consultants and help patients receive appropriate therapy.

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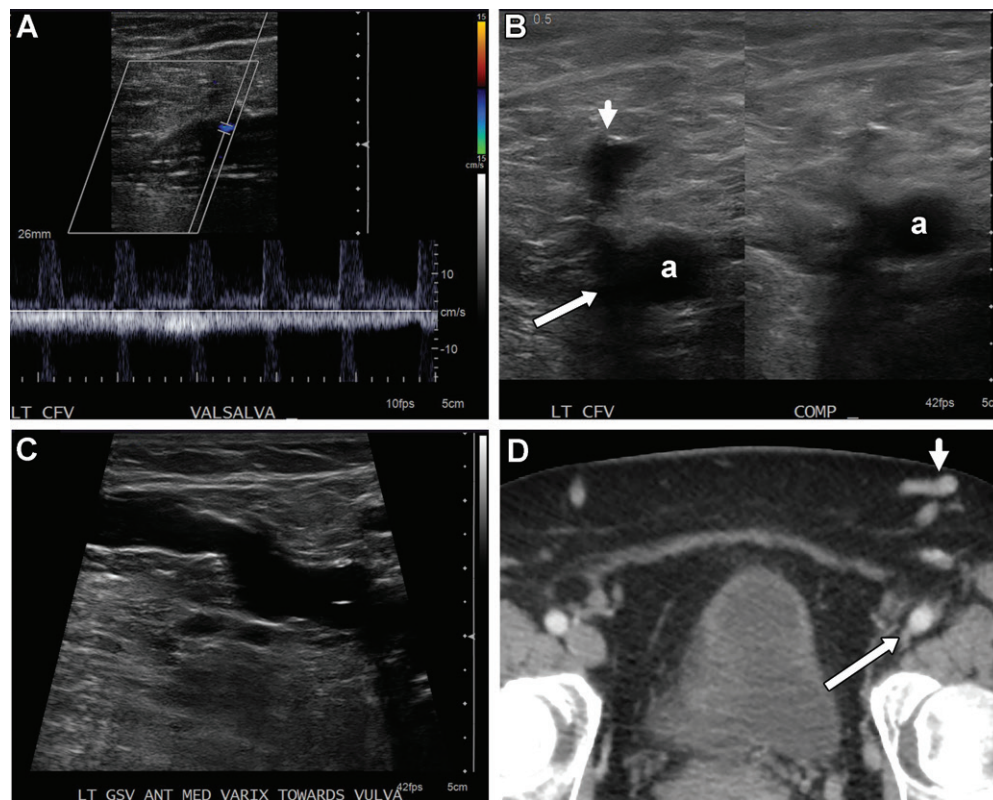


Figure 16. Vulvar varices in a 74-year-old woman. (A) Duplex Doppler US image shows a flat venous waveform in the left CFV without response to the Valsalva maneuver. (B) Dual transverse sonograms of the left CFV without (left) and with (right) compression show a small CFV (long arrow), compared with the adjacent artery (a), somewhat obscured by artifact owing to an overlying superficial vein (short arrow). The relatively larger superficial vein proves to be the superficial external pudendal vein. (C) Oblique US image of the left groin shows the course of the dilated pudendal vein as it courses toward the vulva. (D) Contrast-enhanced CT image shows an atretic scarred left external iliac vein (long arrow) compared with the normal right CFV. The serpentine superficial varix (short arrow) is partially shown in the overlying subcutaneous fat.

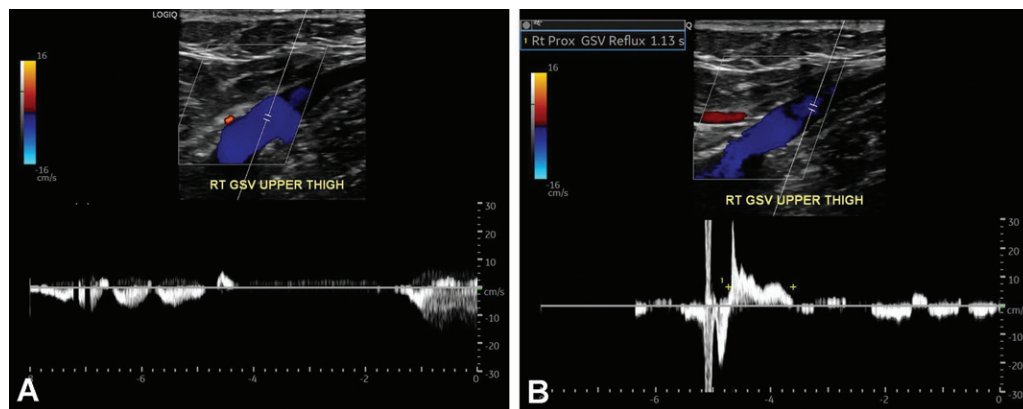


Figure 17. Symptomatic venous insufficiency in a 57-year-old man. (A) Duplex Doppler US image of the GSV during the Valsalva maneuver does not depict reflux. (B) Duplex Doppler US image of the same location in the GSV during an augmentation maneuver shows pathologic reflux in the GSV. When the Valsalva maneuver fails to elicit reflux in a symptomatic patient, augmentation maneuvers are necessary.

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